

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA0308	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
Student Name:	T.Megha Harsha	Reg. No.:	192424389
Section / Batch:	2024	Date:	

Code of Conduct

I, T.Megha Harsha(192424389)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Megha

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

Field	Soil Condition	Irrigation Status	Crop
F1	Dry	Available	Rice
F2	Wet	Available	Wheat
F3	Dry	Unavailable	Maize
F4	Wet	Unavailable	Rice

The system uses the following predicate rules:

- $Dry(x) \wedge IrrigationAvailable(x) \rightarrow NeedsWater(x)$
- $NeedsWater(x) \rightarrow Irrigate(x)$
- $Wet(x) \rightarrow \neg NeedsWater(x)$
- $IrrigationUnavailable(x) \rightarrow \neg Irrigate(x)$
- $Rice(x) \wedge NeedsWater(x) \rightarrow PriorityIrrigation(x)$

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

Search Query	Possible Sense 1	Possible Sense 2
"Amazon tour"	Amazon rainforest	Amazon company
"Safari booking"	Wildlife tour	Web browser
"Java vacation"	Java island	Java programming language
"Cruise package"	Ship journey	Cruise software/project

The system records the following user interactions:

Query	User Interaction
Amazon tour	Viewed rainforest trekking packages
Safari booking	Selected wildlife resort packages
Java vacation	Viewed hotels in Bali and Java
Cruise package	Viewed Mediterranean ship packages

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.

Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

Legal Sentence	Parse Structure	Generated Semantic Interpretation
"The supplier delivered the equipment to the buyer."	Subject–Verb–Object	Supplier = Agent, Equipment = Object, Buyer = Recipient
"The buyer received the equipment from the supplier."	Subject–Verb–Object	Buyer = Agent, Equipment = Object, Supplier = Recipient
"The company terminated the contract after the violation."	Subject–Verb–Object	Company = Agent, Contract = Object
"The contract was terminated by the company."	Passive Construction	Contract = Agent, Company = Object

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation

		structure, visuals, and terminology			and difficult to understand
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Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1	4	3	3	3	3	16	76
2	3	3	3	3	3	15	
3	3	3	3	3	3	15	
4	3	3	3	3	3	15	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Frame Semantics in Banking Customer-Service Systems

1. Analyze the semantic frame of each banking request

A semantic frame represents an event using an action/predicate and its participants or semantic roles.

Query 1

"Transfer ₹20,000 from my savings account to my son's account."

Generated frame:

TRANSFER(SourceAccountAmountDestinationAccountCustomer)

Element	Semantic Role
Transfer	Action
Savings account	Source
₹20,000	Amount
Son's account	Destination
Customer	Agent

Interpretation:

The customer wants to transfer ₹20,000 from the savings account to the son's account.

Query 2

"Block my debit card immediately."

Frame:

BLOCK(DebitCardCustomer)

Element	Semantic Role
Block	Action
Debit card	Object
Customer	Agent

Interpretation:

The customer requests that the debit card be blocked.

Query 3

"Send my transaction statement to my email."

Frame:

SEND(TransactionStatementEmailCustomer)

Element	Semantic Role
Send	Action
Transaction statement	Object
Email	Destination
Customer	Agent

Interpretation:

The customer wants the transaction statement sent to their email.

Query 4

"Increase my daily withdrawal limit."

Frame:

MODIFY(WithdrawalLimitCustomer)

Element	Semantic Role
Increase	Action
Withdrawal limit	Object
Customer	Agent

Interpretation:

The customer wants the withdrawal limit to be increased.

2. Identify the incorrect interpretation

Compare actual intent with system interpretation:

Query	Actual Intent	System Interpretation	Result
B1	Transfer money	Transfer money	Correct
B2	Block debit card	Block debit card	Correct
B3	Receive statement by email	Send statement to email	Correct
B4	Increase withdrawal limit	Decrease withdrawal limit	Incorrect

Therefore:

B4 is incorrect

The customer requested "Increase", but the system interpreted it as "Decrease".

This changes the meaning of the request and could cause an incorrect banking operation.

3. Effect of incorrect semantic-role identification

Incorrect semantic-role identification can be dangerous in banking systems.

For example:

Customer → Increase → Withdrawal Limit

may incorrectly become:

Customer → Decrease → Withdrawal Limit

The system may perform the opposite operation.

Possible consequences:

- Incorrect account modification

- Customer dissatisfaction
- Financial inconvenience
- Security problems
- Loss of trust
- Potential financial loss

Therefore, banking NLP systems should validate the action, object, source, destination, and other semantic roles before executing an operation.

4. Recommended frame-based semantic analysis

A reliable system can use the following architecture:

User Query



Tokenization



POS Tagging



Dependency Parsing



Named Entity Recognition



Semantic Role Labeling



Frame Identification



Action + Participants



Intent Validation



Banking Operation

For example:

"Transfer ₹20,000 from savings to son's account"

TRANSFER

/ | \

Source Amount Destination

Savings ₹20,000 Son's Account

The system should also perform confirmation before high-risk operations:

Detected: Transfer ₹20,000

From: Savings Account

To: Son's Account

Confirm transaction?

Conclusion:

Frame semantics provides a structured representation of banking requests and reduces errors caused by incorrect interpretation of actions and participants.

Q2. First-Order Predicate Calculus for Smart Agriculture

1. Represent field observations using First-Order Predicate Calculus

We define predicates:

- $Dry(x)$
- $Wet(x)$
- $IrrigationAvailable(x)$
- $IrrigationUnavailable(x)$
- $Rice(x)$
- $Wheat(x)$
- $Maize(x)$

Field F1

F1 is dry, irrigation is available and crop is rice.

Dry(F1)

IrrigationAvailable(F1)

Rice(F1)

Field F2

Wet(F2)

IrrigationAvailable(F2)

Wheat(F2)

Field F3

Dry(F3)

IrrigationUnavailable(F3)

Maize(F3)

Field F4

$Wet(F4)$

$IrrigationUnavailable(F4)$

$Rice(F4)$

2. Determine which fields require irrigation

Given rule:

$$Dry(x) \wedge IrrigationAvailable(x) \rightarrow NeedsWater(x)$$

and:

$$NeedsWater(x) \rightarrow Irrigate(x)$$

F1

We know:

$$Dry(F1)$$

and:

$$IrrigationAvailable(F1)$$

Therefore:

$$NeedsWater(F1)$$

Then:

$$Irrigate(F1)$$

Also:

$$Rice(F1) \wedge NeedsWater(F1) \rightarrow PriorityIrrigation(F1)$$

Therefore:

$F1 \text{ requires irrigation and has priority}$

F2

F2 is wet.

Given:

$$Wet(x) \rightarrow \neg NeedsWater(x)$$

Therefore:

$$\neg NeedsWater(F2)$$

Hence:

<i>F2 does not require irrigation</i>

F3

F3 is dry, so it initially suggests water is required.

But:

$$IrrigationUnavailable(F3) \rightarrow \neg Irrigate(F3)$$

Therefore:

$$NeedsWater(F3)$$

but:

$$\neg Irrigate(F3)$$

Hence:

<i>F3 needs water, but cannot be irrigated</i>
--

F4

F4 is wet:

$$Wet(F4) \rightarrow \neg NeedsWater(F4)$$

Therefore:

<i>F4 does not require irrigation</i>

Final result

Field	Water Need	Irrigation	Priority
F1	Yes	Yes	Yes
F2	No	No	No
F3	Yes	No – unavailable	No
F4	No	No	No

3. Should F1 and F3 be treated differently?

Yes.

F1

Dry + Irrigation Available



NeedsWater



Irrigate

Therefore:

$$F1 \rightarrow \text{Irrigate}$$

F3

Dry + Irrigation Unavailable



NeedsWater



Cannot Irrigate

Therefore:

$$F3 \rightarrow \neg \text{Irrigate}$$

The controller should not treat them identically.

F1 can be irrigated immediately, whereas F3 should generate an alert or alternative action because irrigation is unavailable.

4. Is predicate logic alone sufficient?

No.

Predicate logic works well when information is:

- Complete
- Certain
- Clearly defined

However, real agricultural sensors can produce:

- Missing values
- Noisy measurements
- Incorrect readings

- Uncertain soil moisture
- Conflicting sensor information

For example:

Sensor 1: Soil moisture = Dry

Sensor 2: Soil moisture = Wet

Simple predicate logic cannot naturally represent the confidence level of these observations.

A better system can combine predicate logic with:

- Probability
- Fuzzy logic
- Machine learning
- Sensor confidence scores

Example:

$$P(Dry(F1)) = 0.85$$

This represents uncertainty more effectively.

Conclusion:

Predicate logic is useful for rule-based decisions, but real agricultural decision support should combine logical rules with probabilistic or fuzzy reasoning when sensor information is uncertain.

Q3. Word Sense Disambiguation in Travel Recommendation

1. Determine intended senses

"Amazon tour"

Possible meanings:

- Amazon rainforest
- Amazon company

User viewed rainforest trekking packages.

Therefore:

$$\boxed{Amazon = Rainforest}$$

"Safari booking"

Possible meanings:

- Wildlife tour
- Web browser

User selected wildlife resort packages.

Therefore:

Safari = Wildlife tour

"Java vacation"

Possible meanings:

- Java island
- Java programming language

User viewed hotels in Bali and Java.

Therefore:

Java = Java island

"Cruise package"

Possible meanings:

- Ship journey
- Cruise software/project

User viewed Mediterranean ship packages.

Therefore:

Cruise = Ship journey

"Safari download for Windows"

The query contains:

Safari + download + Windows

The user clicked a browser download page.

Therefore:

Safari = Web browser

This demonstrates that the same word can have different senses depending on context.

2. Lexical and contextual cues

Ambiguous Word	Sense	Important Cues
Amazon	Rainforest	tour, trekking, packages
Amazon	Company	shopping, products, delivery
Safari	Wildlife	booking, wildlife, resort
Safari	Browser	download, Windows, browser
Java	Island	vacation, hotels, Bali
Java	Programming	code, programming, developer

Cruise	Ship	Mediterranean, package, journey
Cruise	Software	project, development, application

Two types of clues are important.

Lexical cues

Words directly appearing near the ambiguous word.

Example:

Safari + download + Windows

strongly suggests browser.

User-behaviour cues

The user's actions provide additional evidence.

Example:

Amazon tour

↓

Viewed rainforest trekking

↓

Amazon = Rainforest

3. Why the same word has different interpretations

Words are often polysemous or ambiguous.

For example:

Safari

can refer to:

Safari → Wildlife trip

Safari → Web browser

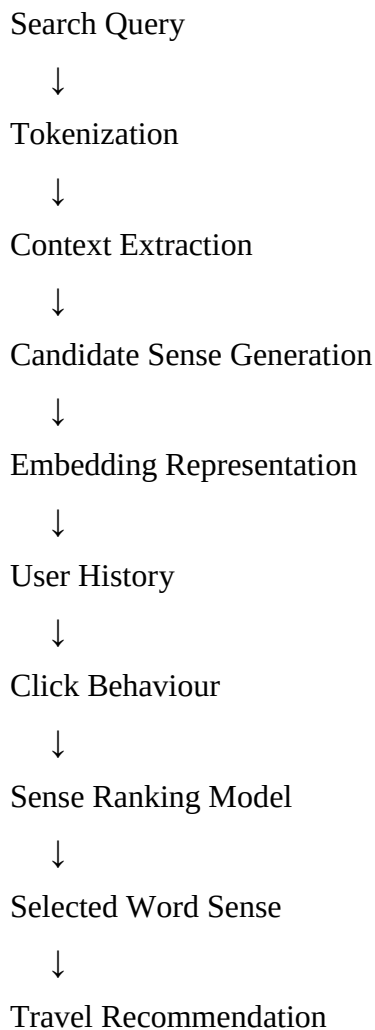
The correct meaning depends on:

- Sentence/query context
- Nearby words
- User history
- Previous searches
- Click behaviour
- Location
- User interests

Therefore, WSD should not rely only on the word itself.

4. Industrial-scale WSD strategy

A practical architecture is:



Step 1: Query context

Analyze surrounding words.

Example:

Safari + download + Windows

→ Browser.

Step 2: Embeddings

Use contextual embeddings such as BERT-like models to represent the meaning of words based on their context.

Step 3: User history

If the user previously searched:

wildlife resorts

jungle tours

Kenya safari

then "Safari" is likely the wildlife sense.

Step 4: Click behaviour

Clicks provide feedback.

If a user repeatedly clicks wildlife packages:

$$P(Wildlife \mid Safari) \uparrow$$

Step 5: Ranking

Combine evidence:

$$Score(s) = w_1 Context + w_2 Embedding + w_3 History + w_4 Clicks$$

Choose the sense with the highest score.

Conclusion:

An industrial WSD system should combine linguistic context with embeddings and behavioural signals to produce personalized and accurate recommendations.

Q4. Syntax-Driven Semantic Analysis in Legal Document Processing

1. How syntax influences semantic roles

Semantic roles identify the function of entities in an event.

Common roles include:

- Agent → performs action
- Object/Patient → receives action
- Recipient → receives something

Active voice

"The supplier delivered the equipment to the buyer."

Structure:

Supplier → delivered → equipment → to buyer

Agent Verb Object Recipient

Therefore:

Supplier = Agent

Equipment = Object

Buyer = Recipient

Passive voice

"The contract was terminated by the company."

Although "contract" is the grammatical subject, it is not the agent.

Contract → was terminated → by company

Object Agent

Therefore:

Contract = Object

Company = Agent

This is why syntax alone must be interpreted carefully.

2. Identify incorrect semantic-role assignments

Sentence 1

"The supplier delivered the equipment to the buyer."

Given:

Supplier = Agent

Equipment = Object

Buyer = Recipient

Correct.

Sentence 2

"The buyer received the equipment from the supplier."

Given:

Buyer = Agent

Equipment = Object

Supplier = Recipient

The first assignment is correct:

Buyer = Agent

But "supplier" is not a Recipient.

"From the supplier" indicates the source of the equipment.

Therefore:

<i>Supplier = Source</i>

not Recipient.

So this interpretation is partially incorrect.

Sentence 3

"The company terminated the contract after the violation."

Given:

Company = Agent

Contract = Object

Correct.

"After the violation" provides temporal/contextual information.

Sentence 4

"The contract was terminated by the company."

Given:

Contract = Agent

Company = Object

This is incorrect.

Correct interpretation:

Company = Agent

Contract = Object/Patient

3. Effect of confusing grammatical subject with semantic agent

This is especially dangerous in legal documents.

Consider:

"The contract was terminated by the company."

Grammatical subject:

Contract

Semantic agent:

Company

If the NLP system assumes:

Subject = Agent

it produces:

Contract = Agent

which is wrong.

This could cause serious errors in:

- Contract analysis
- Liability detection
- Legal event extraction
- Responsibility identification
- Compliance analysis
- Dispute analysis

For example, the system could incorrectly conclude that the contract itself performed the termination.

4. Recommended syntax-driven semantic architecture

A robust legal NLP system can use:

Legal Document



Sentence Segmentation



Tokenization



POS Tagging



Dependency Parsing



Constituency Parsing



Active/Passive Voice Detection



Semantic Role Labeling



Coreference Resolution



Legal Entity Recognition



Semantic Representation



Validation



Legal Information Extraction

Active voice handling

Company → terminated → contract

Agent Object

Passive voice handling

Contract → was terminated → by company

Object Agent

Prepositional phrase handling

For:

delivered equipment to buyer

the system identifies:

equipment → Object

buyer → Recipient

For:

received equipment from supplier

the system identifies:

equipment → Object

supplier → Source

Long-distance dependencies

Legal sentences can be complex:

The contract that the company signed

with the supplier was terminated.

The system should use dependency parsing and coreference resolution to connect related entities even when they are separated by several words.

Conclusion:

A syntax-driven semantic architecture combining dependency parsing, voice detection, semantic role labeling, and coreference resolution can significantly improve semantic interpretation in legal NLP.